Title: Lab 3: Architectural Justification — Student Club Event Ticketing & Budget Portal (CCEM System)

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Architecture Selection: We chose Layered (N-Tier) Architecture for the Student Club Event Ticketing & Budget Portal (CCEM System).

Two Scenario-Related Reasons:

- **Sequential Approval Workflow Isolation:** Layered architecture strictly separates presentation interfaces from business rule processing. This ensures that multi-stage financial budget approvals (Faculty Coordinator -> Finance Office -> Dean) follow their mandated sequential order without risk of client-side bypass.
- **Modular Maintenance & Component Decoupling:** Decoupling high-frequency event ticketing and QR generation from internal administrative budget workflows prevents performance bottlenecks across separate user roles (Club Leads, Approvers, and Venue Staff).

Security Advantage:

- **Role-Based Access Control (RBAC) Enforcement:** Centralizing authorization checks in the “User & Access Management Component” isolates sensitive administrative and financial endpoints. Direct database manipulation is completely blocked from the presentation layer, mitigating unauthorized budget overrides and ticket tampering.

Performance Benefit:

- **Sub-100ms Check-in Latency:** The “Check-in Scanner Component” interfaces directly with optimized caching and database routines. By removing administrative workflow overhead during live events, ticket validation consistently meets the <100ms response requirement under heavy peak load at venue entry gates.